A larger budget, a board that forgets faster or slower

Simulation 2 (asynchronous activation), setup `task003-nosolution`: three variants against the gold standard

LLM-free simulator; report of 19 October 2026; runs of 15, 16 and 18 October 2026

Same sections, figures and equations as the nosolution report of 17 October (itself built like the gold standard), with the four cases

How the simulator works. The world, `task_003`, has three candidate allocations; A0 is the correct one. Each of 24 agents is given one true fact and reasons as an exact Bayesian: its *belief* is the exact probability of A0, A1 and A2 given the facts it currently remembers. There are no rounds. Each agent acts at random times, on average once per time unit: it reads q posts drawn at random from the W most recent posts ($W = 48$ in the gold standard) on a shared *board* (never its own), adds their facts to its memory, recomputes its belief, *votes* by probability matching (a vote drawn from its belief) and posts one remembered fact that supports its vote. Every remembered fact is forgotten after a random lifetime (on average a fraction ρ survives one time unit; reading it again renews it). A *controller* with a target, A0 (truth) or A2 (false), acts at random times at rate λ_c until $t = 30$: it reads q_c recent agent posts and, unless the *vote gate* (at least 75% of them from agents voting for its target) or the *proof stop* (its target already proved; it cannot act in this setup) silences it, posts the one fact from its pool that best supports its target, at most B messages per episode ($B = 30$ in the gold standard). Agents continue alone until $t = 40$. Every episode is run three times with the same random draws: without a controller (*silent*), with the controller targeting A0 and with the one targeting A2.

The facts at the start. All facts are true, and no set of them proves any allocation: the agents' evidence is a coin flip between A0 and A2, and one pool serves both targets (Table 1).

Table 1: Facts available at $t = 0$ and what each set implies, $P(\text{A0}, \text{A1}, \text{A2} \mid \text{facts})$.

set of facts	facts	$P(\text{A0}, \text{A1}, \text{A2})$
agents' facts, all together (24 agents, one fact each)	15 distinct	(0.5, 0, 0.5): no proof
shared pool (both targets)	12, all new to the agents	(0.37, 0.26, 0.37)
agents' facts + pool	27 distinct	(0.75, 0, 0.25)

The three variants. Each changes one rule of the gold standard and keeps everything else (Table 2).

- **Budget.** In the gold standard the controller may post at most $B = 30$ messages before $t = 30$; from $\lambda_c = 2$ upward it uses almost all of them, and at $\lambda_c = 8$ it runs out near $t = 4$. With $B = 100$ it can keep posting until $t = 30$ for $\lambda_c \lesssim 3$ and runs out near $t = 12.5$ at $\lambda_c = 8$; the vote gate can still silence it. This separates the rate from the total effort.
- **Board window.** Nothing is ever deleted from the board, but every reader (agent or controller) draws only from the W newest posts it may read. The 24 agents post about 24 times per time unit, so in the gold standard ($W = 48$) a post stays readable for about 2 time units unless someone posts its fact again. With $W = 24$ it stays about 1 time unit (a board that forgets fast); with $W = 240$ about 10 (a board that forgets slowly). The window also changes what the controller reads, so with $W = 240$ its vote gate sees older votes.

Table 2: The four cases compared in this report. All: $q_c = 12$, $\rho \in \{1, 0.75, 0.5, 0.25\}$, $q \in \{3, 6, 12\}$, all nine rates, gate on and off, 1,000 episodes per setting.

case	budget B	window W	a post stays readable	run
gold standard	30	48	≈ 2 time units	15–16 October (`*_fine`)
$B = 100$	100	48	≈ 2	18 October (`*_budget100`)
$W = 24$	30	24	≈ 1	18 October (`*_window24`)
$W = 240$	30	240	≈ 10	18 October (`*_window240`)

The runs. In this report *gold standard* means the gold-standard rules ($B = 30$, $W = 48$) applied to `task003-nosolution`: the 16 October run behind the 17 October report, at $q_c = 12$ and $\rho \in \{1, 0.75, 0.5, 0.25\}$. The three variants use the same grid: $q \in \{3, 6, 12\}$; rate $\lambda_c \in \{0.5, 1, 2, 3, 4, 5, 6, 7, 8\}$; vote gate on and off; 1,000 episodes per setting (444 settings per case). Unless stated otherwise: $q = 6$, $q_c = 12$, $\lambda_c = 1$, gate on.

Where the budget cannot matter, the $B = 100$ run reproduces the gold standard exactly: all silent settings, and every setting in which the gold-standard controller never reached 30 messages (63 settings).

How this report is organised. Part A (Section 2) compares the gold standard with $B = 100$; Part B (Section 3) compares it with $W = 24$ and $W = 240$. Each part has the gold standard's sections. Only the false controller (target A2) is shown: the controller targeting A0 changes by at most 0.05 in every variant. Readability I and efficiency η still use both targets, as defined in Section 1.

Summary. The agents' evidence is a coin flip between A0 and A2 and nothing can be proved. **The results of the 17 October report hold in all three variants**: the false controller tips the swarm at every memory, and with full memory a faster controller gains less. **The budget does not matter** (Part A): with 100 messages the controller posts two to three times as many and gains at most 0.03 more. **The board's memory matters a little** (Part B): posts readable for 10 time units instead of 2 lower the false controller's gain by 0.02–0.06 when agents forget, posts readable for 1 time unit raise it by up to 0.03. Both effects are much smaller than in `task003-symmetric`.

1 Quantities

Votes. With $n_{a,t}$ the number of agents whose last vote is answer a at time t and N_t the number of agents who have voted,

$$x_{a,t} = n_{a,t}/N_t. \tag{1}$$

At $t = 0$ nobody has voted and we set $x_{a,0} = 0$; all agents have voted by $t \approx 5$. (The 9–14 October reports divided by 24, which counted agents who had not yet acted as voting against the target; this inflated the apparent rise of the shares and the apparent bias of the controller's sensing at early times.)

Paired susceptibility. Following `BLACKBOARD_OBSERVABLES_AND_EFFICIENCIES` (§3), for a controller policy π and an answer a,

$$\chi^{(a)}_{\pi,t} \equiv G^{(a)}_t(\pi) = \mathbb{E}\big[x^{\pi}_{a,t} - x^{\text{base}}_{a,t}\big], \tag{2}$$

the mean over episodes of the difference between the controlled run and the silent run of the same episode. We report it on the controller's own target: $G_t \equiv G^{(0)}_t(\pi^+)$ for the controller targeting A0 and $G_t \equiv G^{(2)}_t(\pi^-)$ for the one targeting A2.

Target and state. The target $Z \in \{\text{A0}, \text{A2}\}$ is drawn by a fair coin. The state X at time t is the vote of a randomly chosen agent, $X \in \{\text{A0}, \text{A1}, \text{A2}\}$. Each episode e of arm s (silent 0, target A0 $+$, target A2 $-$) contributes its vote shares $y^{(e)}_s(x) = x_{x,t}$, and

$$P_s(x) = \frac{1}{E}\sum_{e=1}^{E} y^{(e)}_s(x), \qquad E = 1000. \tag{3}$$

Readability and efficiency. With $Q = \frac{1}{2}(P_+ + P_-)$ and $\mathrm{KL}(p\|q) = \sum_x p(x)\log_2[p(x)/q(x)]$,

$$I(Z;X) = \tfrac{1}{2}\,\mathrm{KL}(P_+\|Q) + \tfrac{1}{2}\,\mathrm{KL}(P_-\|Q), \tag{4}$$

the information the state carries about the target (at most 1 bit). The controllers' total change from silence splits exactly into a directed and an undirected part,

$$C = \tfrac{1}{2}\,\mathrm{KL}(P_+\|P_0) + \tfrac{1}{2}\,\mathrm{KL}(P_-\|P_0) = I(Z;X) + \mathrm{KL}(Q\|P_0), \tag{5}$$

and the efficiency is the directed share,

$$\eta = I(Z;X)/C. \tag{6}$$

Silence sometimes never shows a value of X that a controller reaches, which would make C infinite. Before computing any KL, every arm's distribution therefore receives the same pseudo-count of $\alpha = 0.5$ episodes per value of X (K values),

$$\tilde{P}_s(x) = \frac{E\,P_s(x) + \alpha}{E + \alpha K}. \tag{7}$$

With $\alpha = 1$ instead, η changes by at most 0.004; without it, by at most 0.005 where C is finite. η is not defined when $C < 0.01$ bits (no measurable change); such values are left blank. Equation (5) holds in the data to 4×10^{-15} bits.

Cost. F_t is the number of facts the controller has posted up to t (mean per episode); the gain per fact is G_t/F_t, and the marginal susceptibility is the slope between neighbouring rates,

$$\frac{\Delta G}{\Delta \lambda_c} = \frac{G(\lambda_c^{(k+1)}) - G(\lambda_c^{(k)})}{\lambda_c^{(k+1)} - \lambda_c^{(k)}}. \tag{8}$$

Episode e uses the same random draws at every rate, so the slope is estimated from per-episode differences, $[x^{(e)}_{\text{target},t}(\lambda_c^{(k+1)}) - x^{(e)}_{\text{target},t}(\lambda_c^{(k)})]/(\lambda_c^{(k+1)} - \lambda_c^{(k)})$, whose mean is Eq. (8) and whose standard error gives the 95% intervals (typically ±0.01–0.03, about 30% narrower than for two independent means).

2 Part A: a larger budget ($B = 100$)

The gold standard ($B = 30$) against $B = 100$; the window is $W = 48$ in both.

2.1 Votes over time

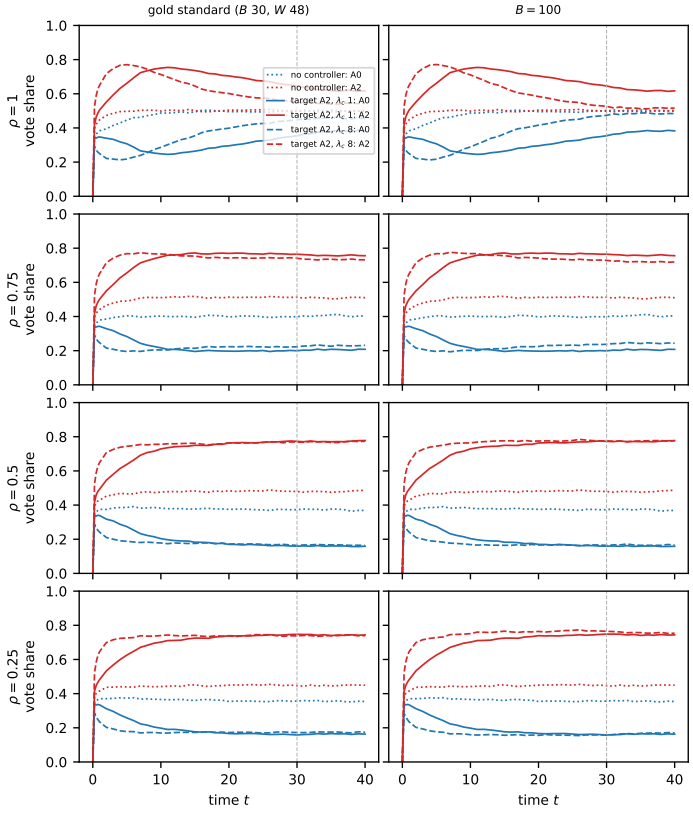


Figure 1: Share of agents voting A0 (blue) and A2 (red), Eq. (1): without a controller (dotted) and with the false controller (target A2) at $\lambda_c = 1$ (solid) and $\lambda_c = 8$ (dashed). Rows: memory ρ. $q = 6$, $q_c = 12$, gate on. At $t = 0$ nobody has voted and every share is 0.

- **The larger budget changes almost nothing**: at $\lambda_c = 8$, A2 on $t = 30$ goes from 0.77 to 0.78 ($\rho = 0.5$) and from 0.55 to 0.53 ($\rho = 1$); at $\lambda_c = 1$ the curves equal the gold standard's. In neither case does the A2 share pass 0.78.

2.2 Susceptibility over time, and per message

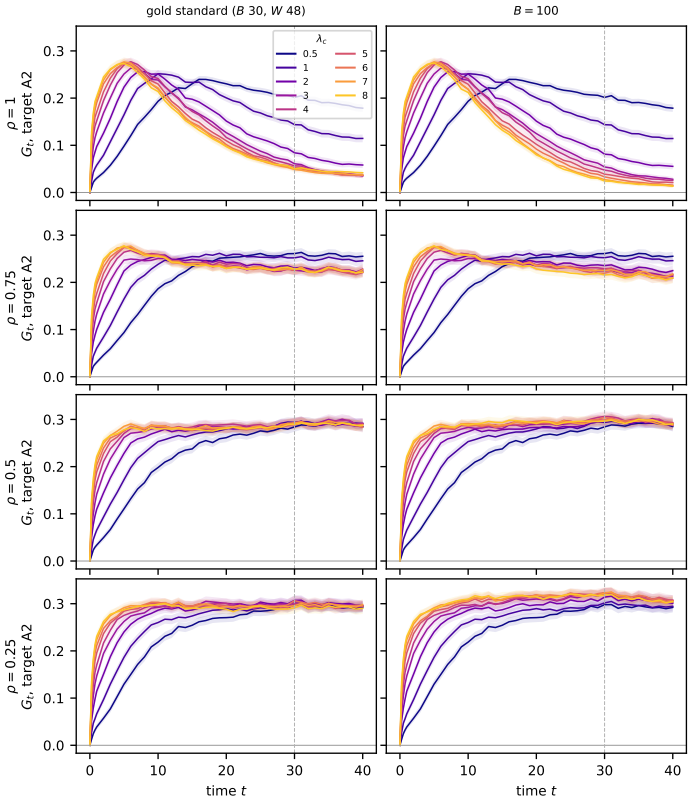


Figure 2: Paired susceptibility G_t, target A2, Eq. (2), one curve per rate λ_c (bands: 95% intervals). Rows: ρ.

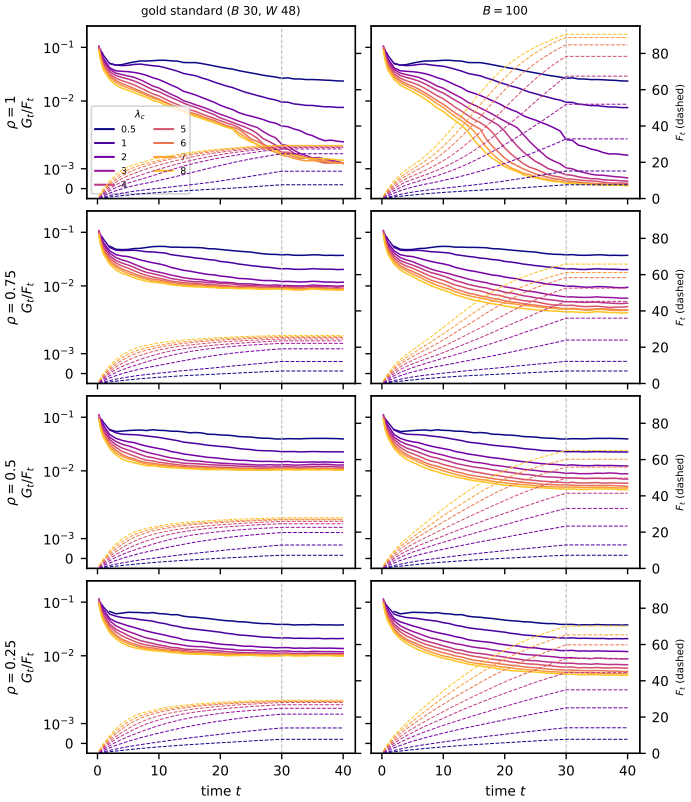


Figure 3: Gain per message G_t/F_t (solid, left axis, log scale) and messages posted F_t (dashed, right axis), target A2, one colour per rate.

- G **moves by at most 0.03** at every rate and memory (Figure 2), although the controller posts two to three times as many messages (at $\lambda_c = 8$: 65–91 instead of 26–29). Fast controllers reach their plateau early (at $\rho = 0.5$, $\lambda_c = 8$: $G = 0.28$ by $t = 5$) and the extra messages come after it.
- **The extra messages are worth almost nothing** (Figure 3): on $t = 30$, $\lambda_c = 8$, the gain per message falls from 0.0104 to 0.0046 at $\rho = 0.5$ and from 0.0017 to 0.0003 at $\rho = 1$.
- **Readability and efficiency are unchanged** (Figure 5): on $t = 30$, $\rho = 0.5$, $I = 0.29$–0.34 bits against 0.29–0.33; $\eta = 0.84$–0.92 against 0.86–0.92 at $\rho < 1$.

2.3 Phase diagrams

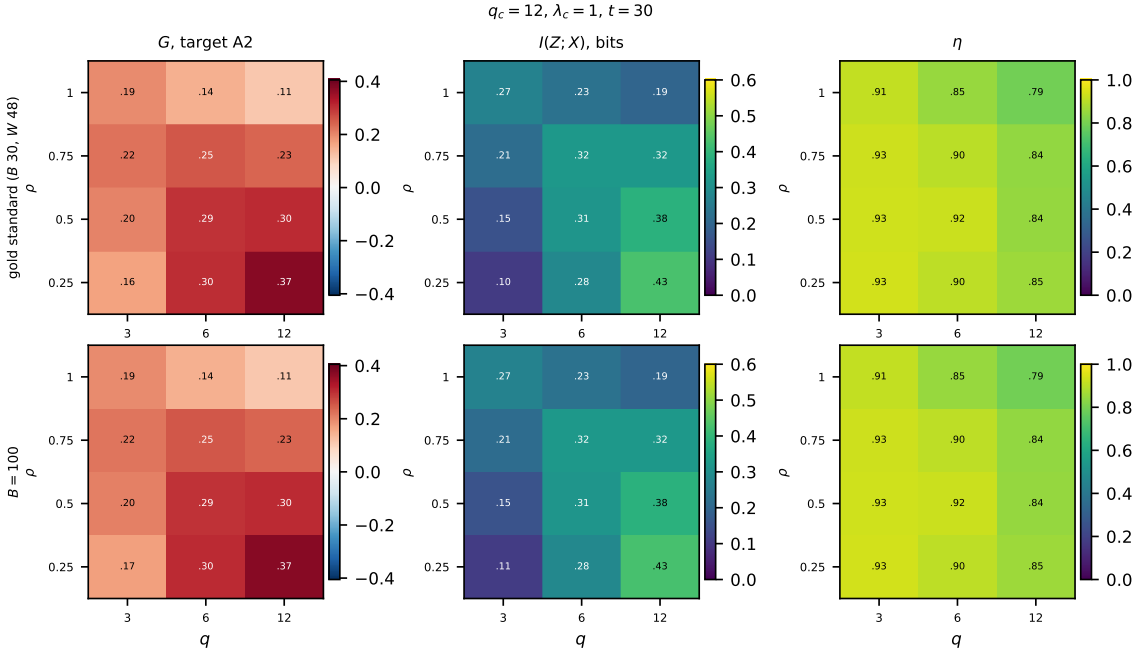


Figure 4: Susceptibility (target A2; grey number: 95% interval includes 0), readability and efficiency on $t = 30$ over memory ρ and the posts each agent reads q. $q_c = 12$, $\lambda_c = 1$, gate on.

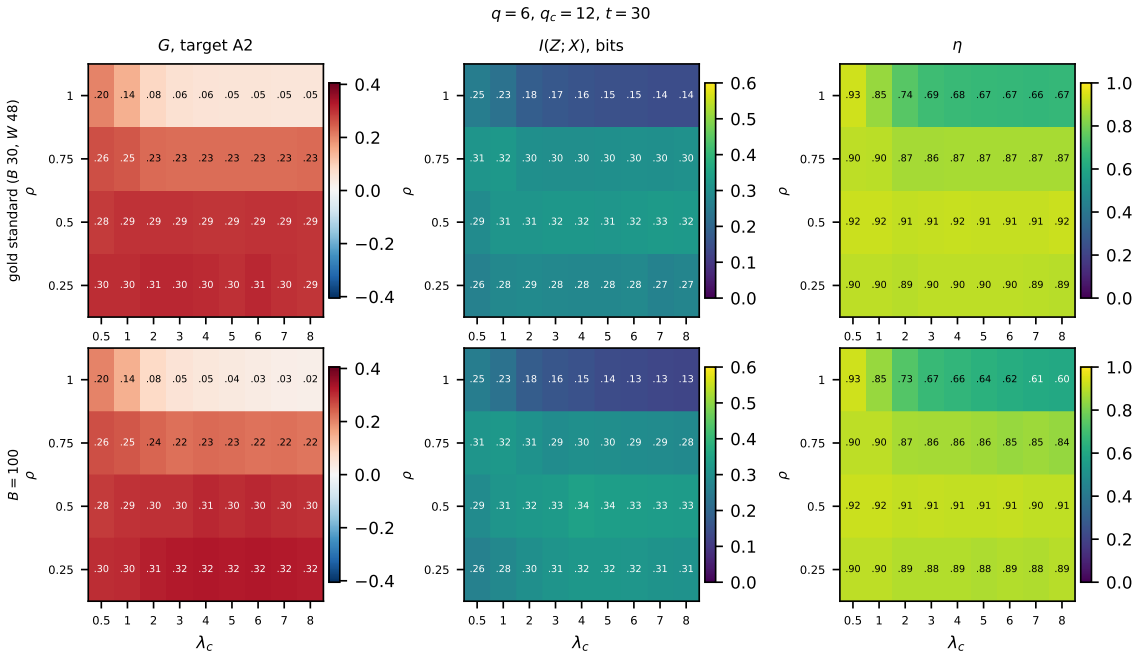


Figure 5: The same over memory ρ and the rate λ_c. $q = 6$, $q_c = 12$.

- **The reading reversal is unchanged** (Figure 4): from $q = 3$ to 12, G falls with full memory (0.19 → 0.11 in both) and rises with heavy forgetting (0.16 → 0.37 gold standard, 0.17 → 0.37 $B = 100$).
- **The rate still hardly matters at** $\rho < 1$ (Figure 5): from $\lambda_c = 1$ to 8, G changes by at most 0.03 in both.

2.4 Susceptibility against the rate, against messages, and its slope

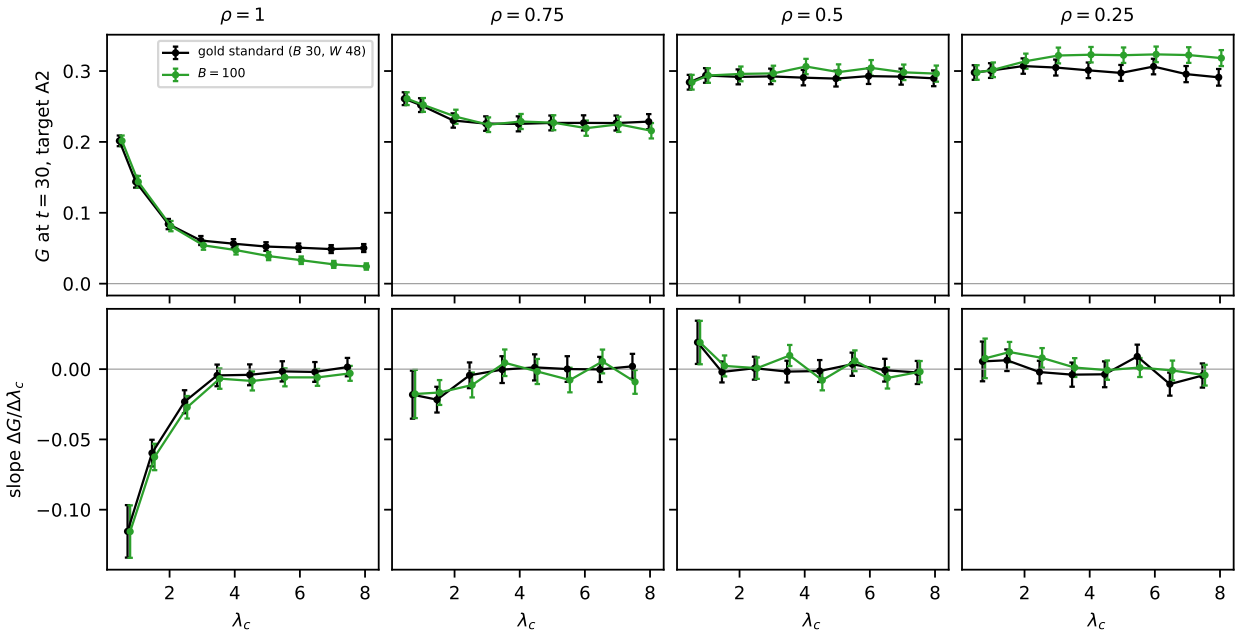


Figure 6: Top: G on $t = 30$, target A2, against the rate (bars: 95% intervals). Bottom: the slope between neighbouring rates, Eq. (8), with paired 95% intervals. $q = 6$, $q_c = 12$, gate on.

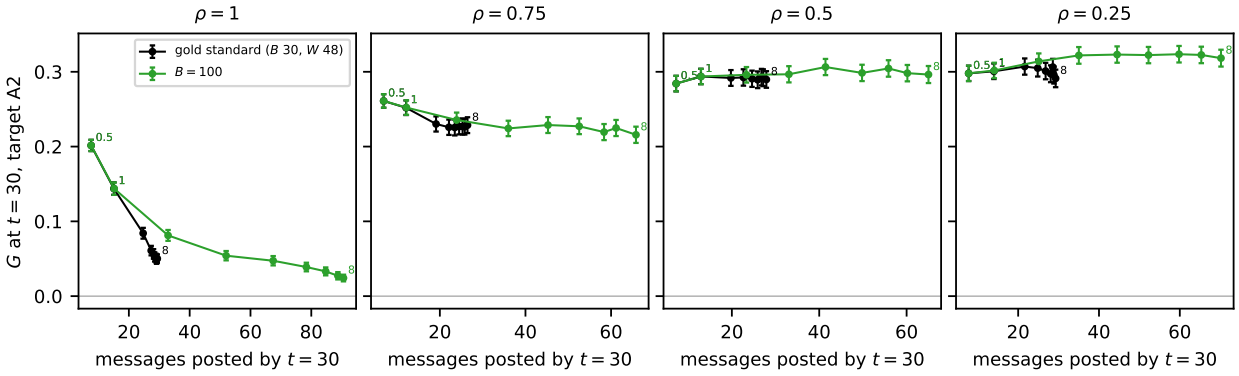


Figure 7: G on $t = 30$, target A2, against the messages posted by then; one point per rate, joined in rate order (labels: $\lambda_c = 0.5$, 1, 8); bars: 95% intervals. Moving right is more messages, moving up is more gain.

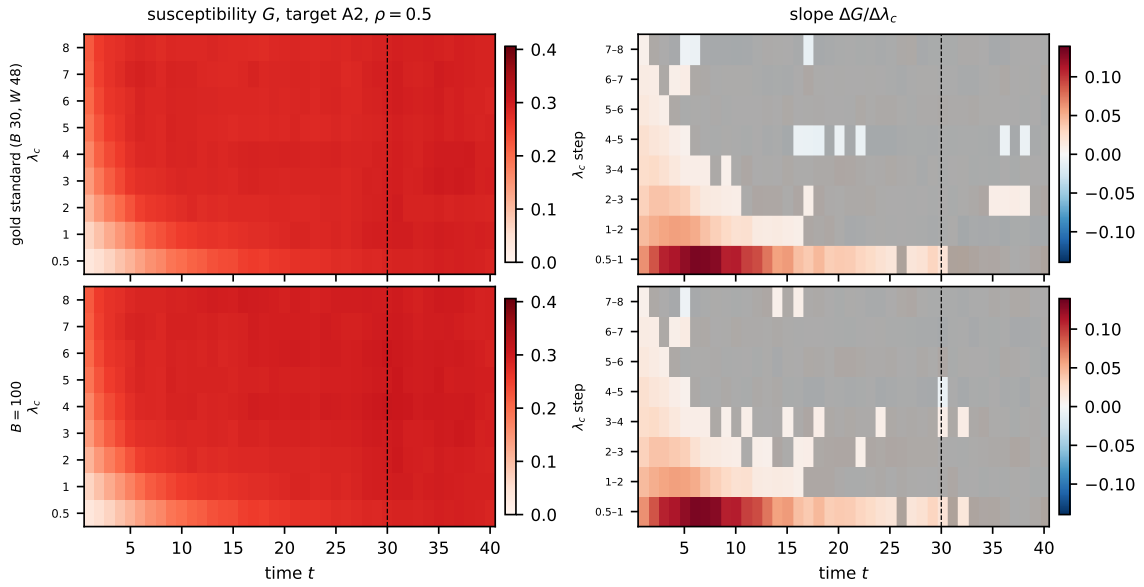


Figure 8: G, target A2 (left), and its slope between neighbouring rates (right) over time and rate, $\rho = 0.5$. Faded cells: the paired 95% interval of the slope includes 0.

- **More messages do not raise the gain** (Figure 7): with $B = 100$ the curves stretch to the right but stay flat; at $\rho = 0.5$ the false controller gains 0.28–0.31 whether it posts 13 or 65 messages.
- **The rate reversal at full memory is there in both** (Figure 6): the first steps of rate cost -0.12 and -0.06 per unit; from $\lambda_c = 0.5$ to 8, G falls from 0.20 to 0.05 (gold standard) and to 0.02 ($B = 100$).
- **Over time the slopes are the same** (Figure 8): at $\rho = 0.5$ more rate helps early (on $t = 5$ the first step is worth $+0.12$ per unit) and that advantage fades to $+0.02$ by $t = 30$, in both.
- Gate off minus on: -0.08 to $+0.03$ (gold standard), -0.08 to $+0.01$ ($B = 100$), the largest at $\rho = 1$.

What the budget changes. Nothing measurable: the controller with 100 messages posts two to three times as many and gains at most 0.03 more. A likely reason, not tested: without a proof, an agent's belief is limited by what the facts imply, and once agents hold the controller's facts, repeating them adds nothing.

3 Part B: a board that forgets faster or slower ($W = 24$, $W = 240$)

$W = 24$ (posts readable for about 1 time unit), the gold standard ($W = 48$, about 2) and $W = 240$ (about 10); the budget is $B = 30$ in all three.

3.1 Votes over time

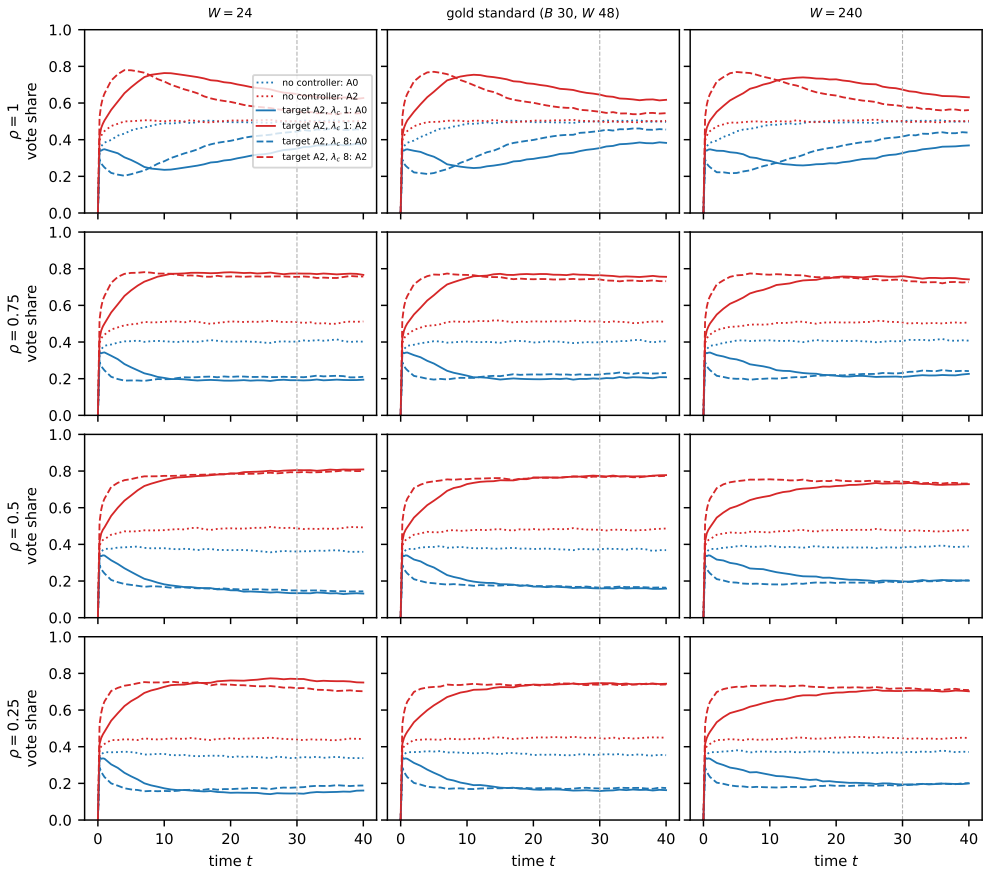


Figure 9: As Figure 1, for the three windows.

- **Without a controller the window hardly matters**: the shares on $t = 30$ differ by at most 0.02.
- **The false controller tips the swarm with every window**: A2 on $t = 30$ at $\lambda_c = 1$ is 0.81, 0.77 and 0.73 for $W = 24$, 48, 240 at $\rho = 0.5$; at $\rho = 1$, 0.65, 0.65 and 0.67.

3.2 Susceptibility over time, and per message

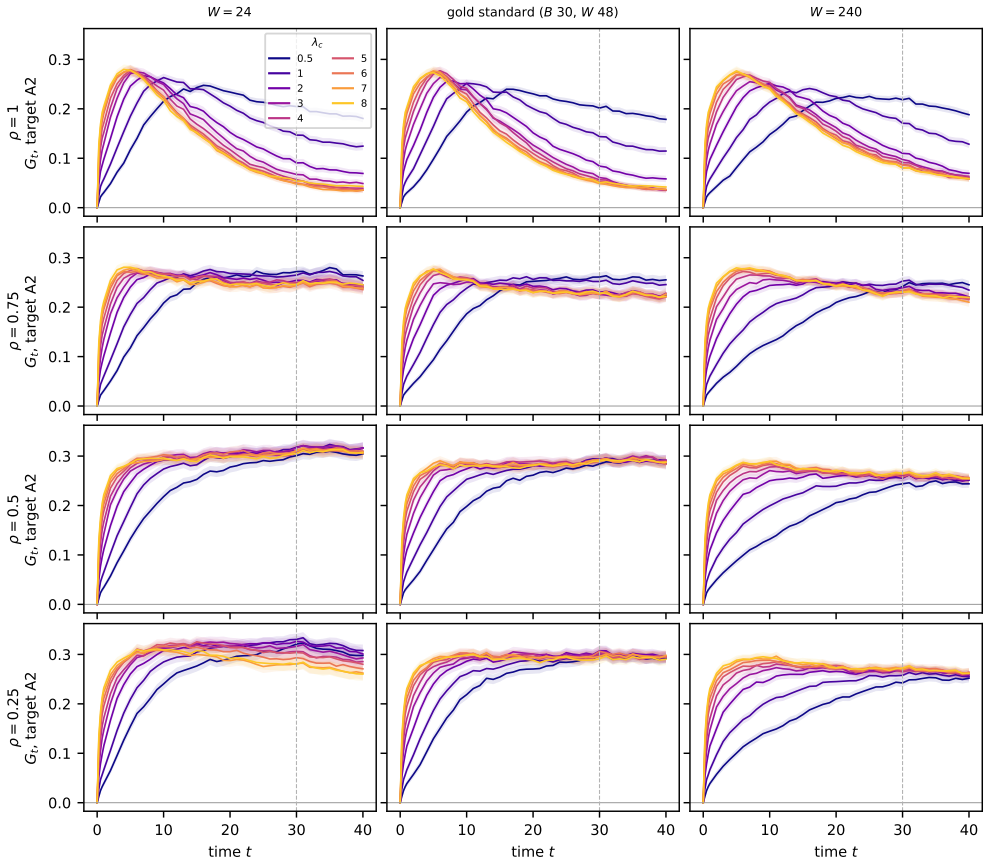


Figure 10: Paired susceptibility G_t, target A2, one curve per rate (bands: 95% intervals).

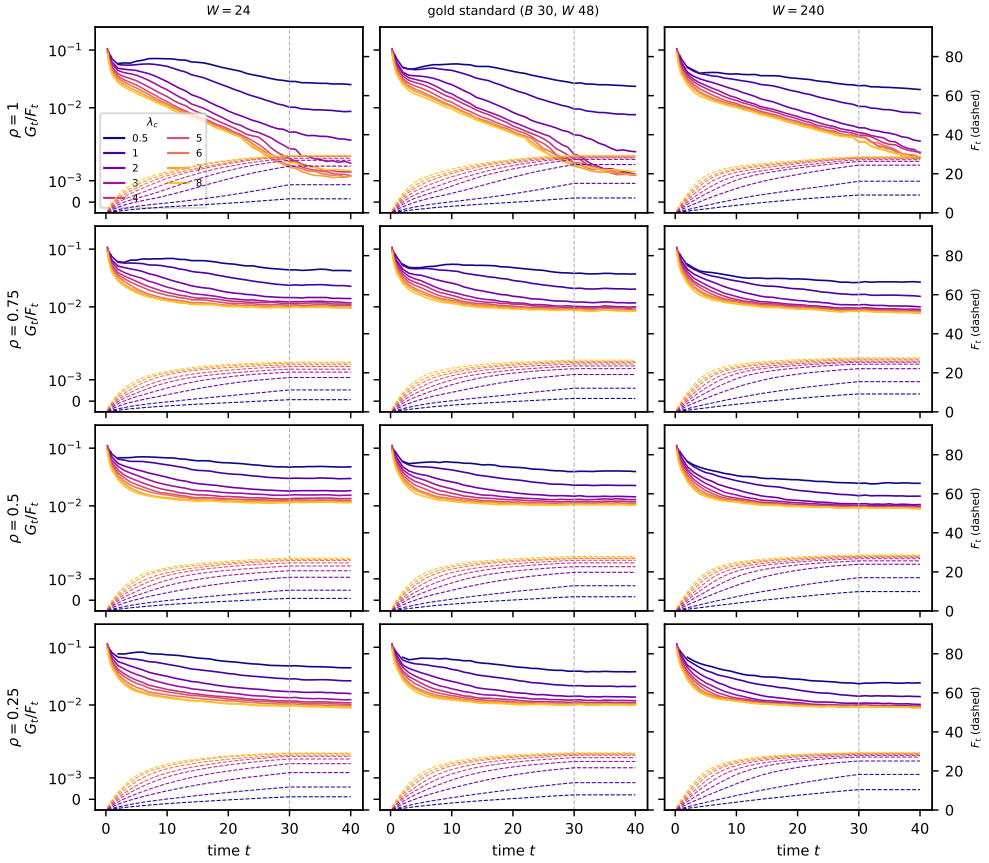


Figure 11: Gain per message G_t/F_t (solid) and messages posted F_t (dashed), target A2, one colour per rate.

- **The window moves G a little** (Figure 10): rate by rate, $W = 24$ by -0.01 to $+0.03$; $W = 240$ by -0.02 to -0.06 at $\rho \leq 0.5$.
- **At full memory $W = 240$ helps the false controller** ($+0.02$ to $+0.04$) and weakens the rate reversal: G falls from 0.22 ($\lambda_c = 0.5$) to 0.08 ($\lambda_c = 8$), against 0.20 to 0.05.
- **Each message is worth more when the board forgets fast** (Figure 11): at $\lambda_c = 1$, $\rho = 0.5$: 0.030, 0.023 and 0.015 per message.
- **Readability follows, efficiency does not move** (Figure 13): on $t = 30$, $\rho = 0.5$, $I = 0.33$–0.37, 0.29–0.33, 0.21–0.26 bits; $\eta = 0.85$–0.92 at $\rho < 1$ for every window.

3.3 Phase diagrams

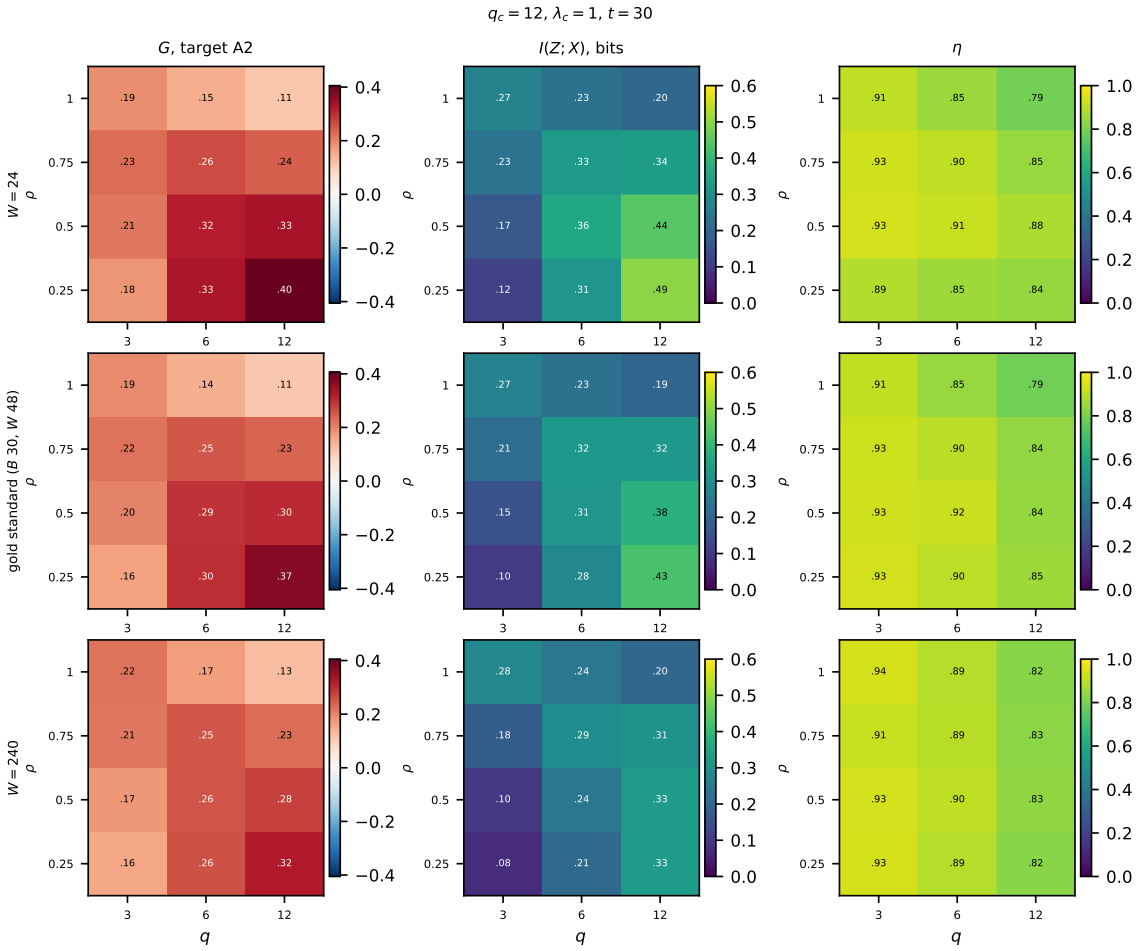


Figure 12: As Figure 4, for the three windows.

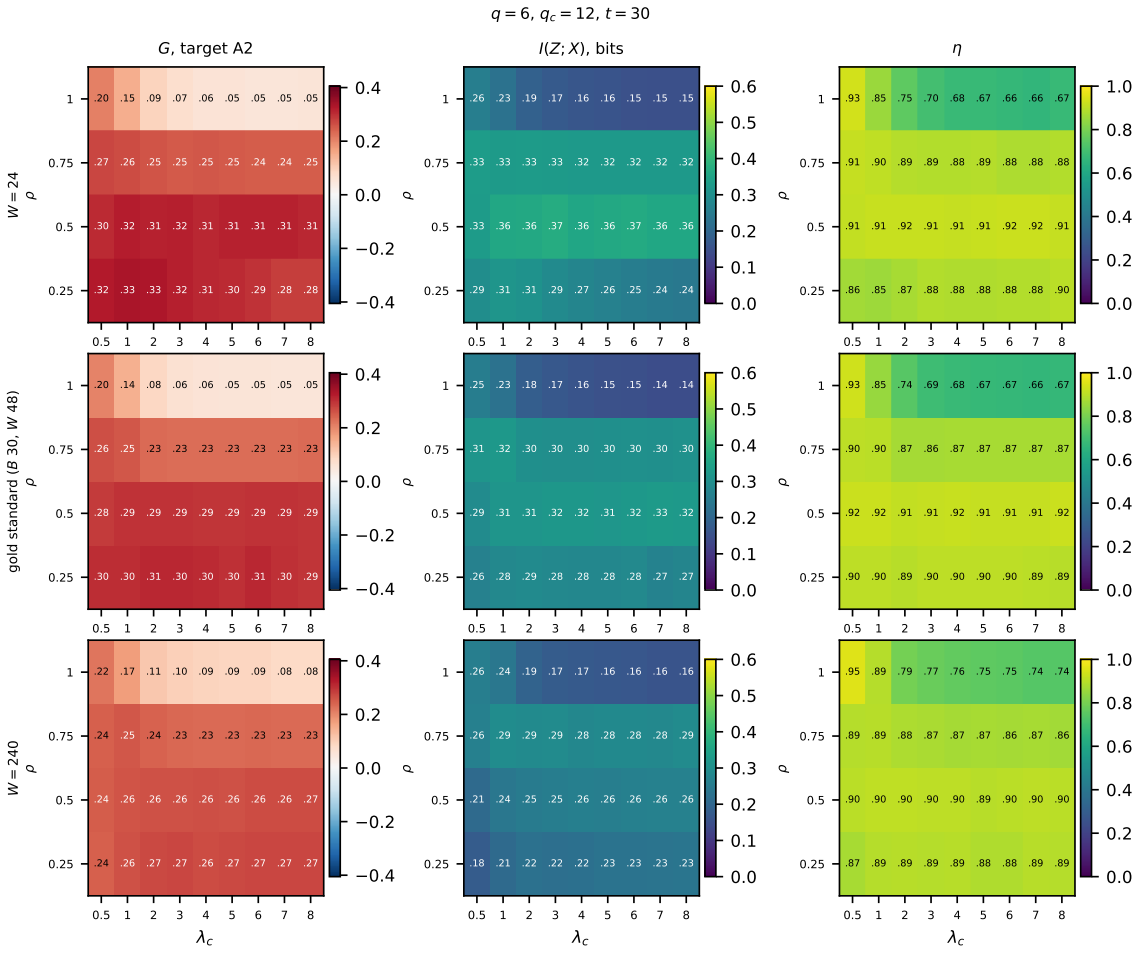


Figure 13: As Figure 5, for the three windows.

- **The reading reversal survives every window** (Figure 12): from $q = 3$ to 12, G falls with full memory (0.19 → 0.11, 0.19 → 0.11, 0.22 → 0.13 for $W = 24$, 48, 240) and rises with heavy forgetting (0.18 → 0.40, 0.16 → 0.37, 0.16 → 0.32).
- **The rate hardly matters at $\rho < 1$ with any window** (Figure 13): from $\lambda_c = 1$ to 8, G changes by at most 0.05.

3.4 Susceptibility against the rate, against messages, and its slope

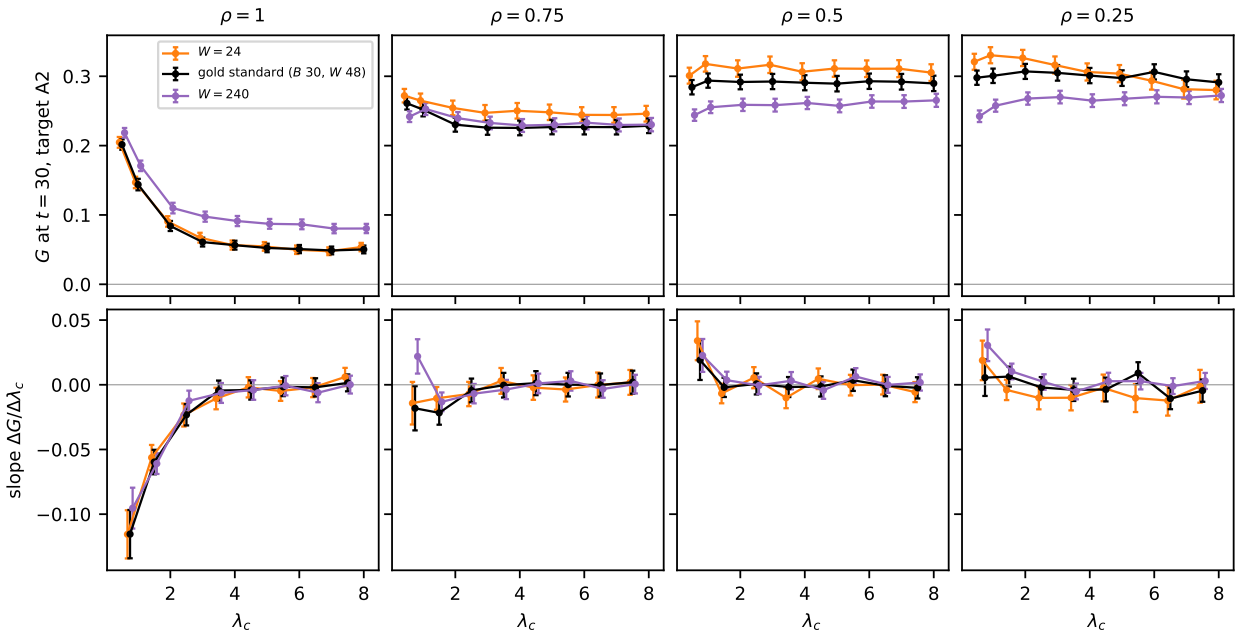


Figure 14: As Figure 6, for the three windows.

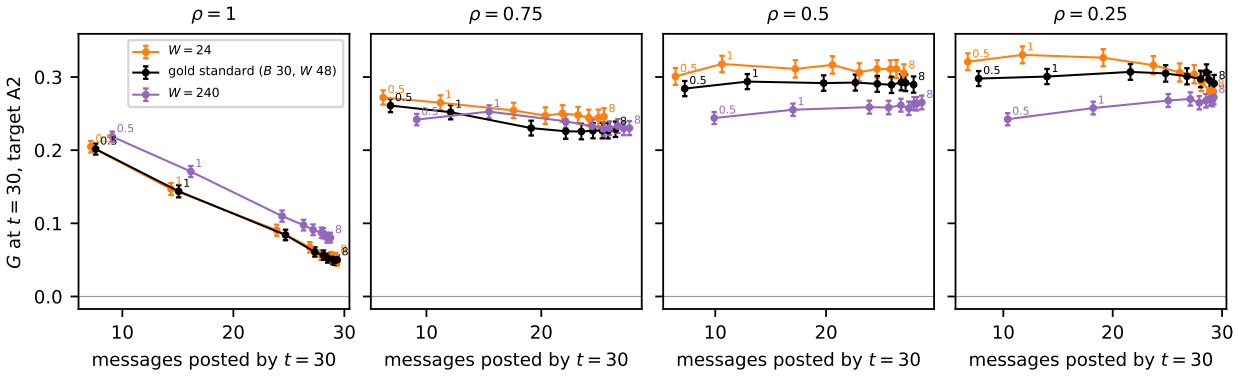


Figure 15: As Figure 7, for the three windows.

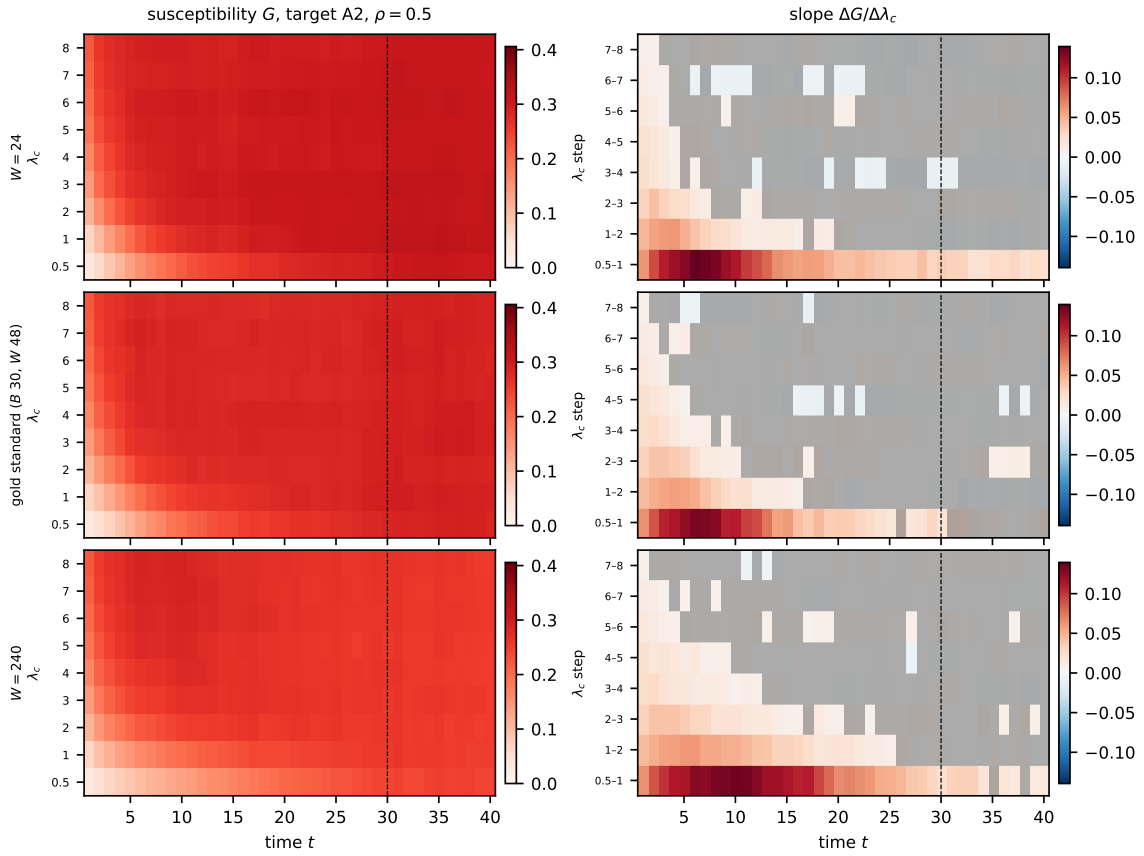


Figure 16: As Figure 8, for the three windows.

- **The rate reversal at full memory is there with every window** (Figure 14): the first step of rate costs -0.12 ($W = 24$, 48) or -0.10 ($W = 240$) per unit, the second -0.06. At $\rho < 1$ every significant slope on $t = 30$ is small (≤ 0.034).
- **With $W = 240$ the early advantage of speed lasts longer** (Figure 16): at $\rho = 0.5$ the first step is still worth $+0.07$ per unit on $t = 20$, against $+0.04$ for the other windows.
- Gate off minus on: between -0.08 and $+0.03$ for every window, the largest at $\rho = 1$.

No proofs here. In `task003-nosolution` no set of facts proves any allocation, so the gold standard's "who holds the proof" section has no counterpart in either part.

What the window changes. A little: a longer window lowers the false controller's gain by 0.02–0.06 when agents forget, a shorter one raises it by up to 0.03; with full memory a longer window helps the false controller slightly and weakens the rate reversal.

4 Conclusions

1. **The results of the 17 October nosolution report hold in all three variants**: the false controller tips the swarm at every memory, full memory included; with full memory a faster controller gains less; reading more helps the controller when agents forget and hurts it when they remember; η stays at 0.84–0.92 for $\rho < 1$.
2. **The budget does not matter here** (Part A), and **the window matters a little** (Part B).
3. **Compared with `task003-symmetric`**: there, a larger budget raised the false controller's gain by up to 0.13 and the window moved it by up to 0.14; here both effects are at most 0.06. Without a proof to erode, the swarm saturates after a few messages.
4. **The controller targeting A0** is left out of this report: in every variant its gain on $t = 30$ stays within 0.05 of the gold standard's.

Caveats. The window changes what the controller reads too: with $W = 240$ its vote gate sees votes up to 10 time units old, and at $\lambda_c = 1$ it posts 15–18 messages instead of 12–15. The variants were run at $q_c = 12$ and four memory values only, so the gold standard's $q_c \times \lambda_c$ diagram has no counterpart. One world, 24 exact-Bayesian agents, one controller policy; 1,000 episodes per setting (95% intervals on G about ± 0.02). Data: `results/simulation_2/2026-10-18_sim2_nosolution_*`; tables: `results/gold_standard/2026-10-18/` (variants) and `2026-10-16/` (gold standard); code: `shared_analysis/gold_standard/make_report_figures_2026-10-19.py`.